

EXERCISES AND EXACT SOLUTIONS

Samples That Can Represent a Nation

Independent practice, facilitated discussion, and assessment reference

Production Unit	PU-B07-C17
Notebook	V7_A_N12
Version	0.1
Assessment boundary	Learning evidence—not operational authorization

Instructions

Complete Part A before consulting the solutions. State assumptions explicitly. Where evidence is insufficient, write an abstention or escalation response rather than inventing precision.

Exercise 1

Explain why weights are required.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 2

Add cluster design effects.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exercise 3

Design nonresponse adjustment.

- Required output: a concise decision-ready response with owner, evidence, uncertainty, and boundary where applicable.

Exact solutions and scoring guide

Solution 1

Unequal inclusion probabilities require inverse-probability weights for population inference.

Solution 2

Inflate the effective sample using justified intracluster correlation and cluster size assumptions.

Solution 3

Define response cells or a response-propensity model using pre-outcome variables, cap extreme adjustments, and validate bias/variance.

Scoring rubric

Criterion	Strong evidence	Revision required
Decision framing	Owner, action, horizon, authority named	Model output substituted for decision
Evidence	Concepts, grain, provenance, limits explicit	Data treated as self-explanatory
Uncertainty	Abstention/escalation used appropriately	False precision or hidden limitations
Governance	Human authority and safeguards retained	Authority silently delegated